ADAGIO GROUP

Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Review #2: Instantaneous Present

Benjamin D. Summers <bsummers@adagiollc.com> Mon, Sep 7, 2026 at 6:12 AM
To: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>
Cc: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Hi Sara,

Thank you for the review. The revised manuscript is attached. Below is what I did with each point and why, starting with the one that explains most of the others...

This paper was written before the subsequent papers on indivisible processes, and its claim is that a complete dynamical state at a temporal point is not physically justified. After consecutive desk rejections I revised the title, the abstract, and the Discussion toward the indivisibility program, and by the version you read the paper was claiming a result about Chapman-Kolmogorov divisibility that the three arguments don't deliver. The closing paragraph of the Discussion contradicted the introduction, and I didn't see it until your report set the two claims next to each other. The revision restores the original claim under the title "The Physics of the Instant: Analysis from Limits, Relativity, and Renormalization." Chapman-Kolmogorov now appears once, in Section 6, and the only thing said about it is that it inherits the assumption.

1. Operational versus ontological. Accepted. The paper no longer claims that the state at a point is incomplete. It claims that completeness at a point is an assumption, that the construction of the state doesn't supply it, and that established physics gives no basis for it and positive reasons to doubt it. Section 3 says outright that the operational result does not establish incompleteness of the mathematical state. That is your moderated conclusion, with the two additions I flag in item 12.

2. Abstract. Accepted; rewritten. "Guarantees" is now "shows," with the quantum-gravity caveat stated in the abstract. "Indivisibility is the generic expectation" and "quantum behavior is the default" are gone. On the second: generic indivisibility gives memory; quantum processes are the unistochastic subclass; I had run the class and the subclass together.

3. Section 2, the limit. Accepted. You're right that the punctured neighborhood is there to make the limit independent of a separately assigned f(a), and that L has whatever structure the state space has. The section now concedes both, opens by acknowledging that theories introduce point-indexed states directly, and puts the assumption where it belongs: in continuity in time, which identifies the value at the point with the limit of the approach, and in the dynamical sufficiency of that value. Neither comes from the definition of a limit, and that is now all the section claims.

4. Section 2, Landau-Lifshitz. Accepted; the paragraph is out. Brownian motion is Markov with no velocity, so the derivative point can't carry the completeness point. What replaces it is the argument I should have made in the first place: a function of a Markov process is generically not Markov (Burke and Rosenblatt 1958; Kemeny and Snell 1960), so any process observed through a finite window is non-Markov at every resolution, and the composition defect vanishes only as the window closes. Divisibility is a property of the limit and of nothing along the way to it. I checked this in exact rational arithmetic on a two-state chain: the fine chain's defect is identically zero; the windowed family's defect is nonzero at every width tested and scales linearly with the width as the width goes to zero.

5. Section 3, Mead-Garay and 't Hooft. Accepted. The section now opens with the caveat you asked for: the bounds come from extrapolating both theories past their verified domains, they are robust across the approaches in Garay's review, and they are not theorems of quantum gravity. Minimum measurable resolution is distinguished from discreteness and from the failure of point-indexed states. 't Hooft's result is confined to the scattering configuration.

6. Section 3, Salecker-Wigner. Accepted, and this was the worst error in the text: the paragraph claimed independence from gravitation and then used the Schwarzschild radius. The quantum bound $M \gtrsim \hbar T/(c^2 \delta t^2)$ now stands alone; the collapse condition and $\ell_clock \lesssim c\delta t$ are stated as the additional assumptions; the combination gives $\delta t \gtrsim (T\ t_P^2)^{\{1/3\}}$, Károlyházy's and Ng and van Dam's form, with $\delta t \gtrsim t_P$ as the case $T \sim \delta t$. Independence from Mead and 't Hooft is claimed only in that the argument constrains the clock, not the region. Checking the bound also turned up that its T-dependence is disputed (Diósi and Lukács 1996; Baez and Olson 2002; Ng and van Dam replied). The text says so and uses only the Planck-scale corollary.

7. Section 4, renormalization. Accepted. "Physics at a point is infinite" is now "products of field operators at coincident points are singular." Continuum limits are described as taken on renormalized quantities and as theory-dependent: lattice φ^4 in four dimensions has a Gaussian scaling limit (Aizenman and Duminil-Copin 2021); asymptotically free theories are the interacting case. The lattice sentence now says what you said, finite at fixed spacing, with

renormalization entering across scales and in the continuum limit. The claim that the bare theory at a point is "not a physically meaningful object" is gone; its status is now settled theory by theory.

8. Section 4, algebraic QFT. Accepted. You asked for the assumptions behind the point-algebra statement; I could not source that statement to a locus, so it is replaced by what can be sourced: the Haag-Kastler assignment of observables to open regions, and Wightman's 1964 result that pointwise-defined fields lead to contradictions under natural assumptions. Both are limited to the localization of observables, with the explicit statement that they do not bear on the composition of the dynamics, which in that framework sits in the automorphism group.

9. Section 5. Accepted. The section is retitled "The Collapse-Timing Formulation of the Measurement Problem" and names that formulation in its first paragraph. The sentence claiming the general conflict has no well-defined terms is out; it asserted what the next sentence disclaimed. Decoherence is cited only for the finite timescale of entanglement. CSL (Ghirardi, Pearle, and Rimini 1990; Bassi et al. 2013) is cited to show that the tension does not depend on an instant. The scope of Barandes's correspondence, finite configuration spaces and unistochastic matrices, is stated here and in Section 6.

10. Section 6. Accepted. The δt_obs ≫ τ_coh paragraph is deleted; it had no model behind it, and you're right that coarse-graining can create memory as easily as remove it. The prediction paragraph is replaced by the one hypothesis the window-averaging result supports: for a system that is Markov on the observed state space, the composition defect must fall to zero as the window is refined, and a defect that persists under refinement is memory in the chosen state space, not in the resolution. The Motahari citation is removed; with the prediction gone there is nothing for it to support.

11. Section 7 and references. Accepted. The last paragraph is rewritten as a claim about an assumption, consistent with everything above it. The divisibility and process-tensor literature (Breuer, Laine, and Piilo 2009; Rivas, Huelga, and Plenio 2010; Milz and Modi 2021) is now cited as the starting point for the criteria the paper places out of scope; the quantitative treatment stays in the companion papers, which this paper no longer cites.

12. I don't fully agree with two points: 1) Your report says the absence of a guarantee is not by itself evidence that completeness fails; the paper says established physics gives positive reasons to doubt it, and rests that on Sections 3 and 4, where the point is not merely unguaranteed but unreachable and singular. 2) Your moderated conclusion stops at divisible models being an idealization to be assessed system by system; the paper goes one step further and says the burden of justification is on the memoryless description. I've kept both, the second as an argued position built on items 4 and 10, not as something the three physical arguments prove. If you think either step is still unearned, tell me where it fails.

Bibliography: Landau-Lifshitz and Motahari are out; in are Burke-Rosenblatt, Kemeny-Snell, Károlyházy, Ng-van Dam, Diósi-Lukács, Baez-Olson, Aizenman-Duminil-Copin, Ghirardi-Rimini-Weber, Ghirardi-Pearle-Rimini, Bassi et al., Breuer-Laine-Piilo, Rivas-Huelga-Plenio, Milz-Modi, and Wightman 1964. The companion papers cite this one under the old title; that gets fixed at their next revisions.

Let me know your thoughts.

Ben

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On Sun, Sep 6, 2026 at 7:42 AM ISMAIL-SUTTON, SARA R. <sara.r.ismail-sutton@durham.ac.uk> wrote:

> Dear Ben
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> Please find attached the review of "The Incompleteness of the Instantaneous State and the Status of Chapman-Kolmogorov Divisibility".
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> Best regards
> Sara
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> **From:** Benjamin D. Summers <bsummers@adagiollc.com>
> **Sent:** 04 September 2026 18:43
> **To:** ISMAIL-SUTTON, SARA R. <sara.r.ismail-sutton@durham.ac.uk>
> **Subject:** Re: Review #2 Re: Koopman Spectral Theory Paper